

Literature Status for arXiv:math/0505302: State-Preserving CP Approximation

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Status

This is a `literature_already_answered` packet, not a new counterexample of this run. Eckhardt [2] explicitly answers a question of Ricard and Xu [1] by constructing a faithful state on a nuclear unital C^* -algebra which has no state-preserving finite-rank UCP approximation of the identity.

The source question

In the CCAP application section of [1], the authors ask whether every nuclear unital C^* -algebra A , equipped with a given state ϕ , possesses a net of finite-rank, unital completely positive maps $T_\alpha : A \rightarrow A$ such that

$$\phi \circ T_\alpha = \phi \quad \text{and} \quad \|T_\alpha(a) - a\| \longrightarrow 0 \quad (a \in A).$$

Their reduced-free-product theorem shows why this is relevant: if every factor state has such approximants (together with the stated faithful GNS hypotheses), then the reduced free product has the completely contractive approximation property.

Eckhardt's negative answer

Let m be Lebesgue measure and choose a measurable set $X \subset [0, 1]$ such that, for every nonempty open set $\mathcal{O} \subset [0, 1]$,

$$m(X \cap \mathcal{O}) > 0, \quad m(X^c \cap \mathcal{O}) > 0.$$

On the nuclear unital algebra $A = M_2 \otimes C[0, 1]$, Eckhardt defines

$$\phi((f_{ij})) = \int_X f_{11} dm + \int_{X^c} f_{22} dm. \tag{1}$$

The two-sided positive-measure property makes ϕ faithful.

Proposition 2.3 of [2] proves the decisive rigidity statement. After the paper's preceding corner-preserving reduction, write a finite-rank, ϕ -preserving UCP map as

$$S = \begin{pmatrix} S_{11} & S_{12} \\ S_{12}^* & S_{22} \end{pmatrix}.$$

Then necessarily $S_{12} = 0$. The proof combines positivity of 2×2 operator matrices with local averaging on intervals; the defining property of X forces the off-diagonal coefficient to vanish pointwise.

If a net of such maps converged pointwise to the identity, its action on the off-diagonal corner $e_{12} \otimes C[0, 1]$ would converge to that corner. But every reduced approximant annihilates it. This contradiction proves that the faithful state (1) is not CP-approximable. The abstract of [2] explicitly identifies this as an answer to Ricard–Xu.

Scope

The source’s auxiliary state-preserving approximation question is therefore false, even for a faithful state on the elementary nuclear algebra $M_2 \otimes C[0, 1]$. This does *not* disprove CCAP stability under arbitrary reduced free products: the state-preserving approximation condition was a sufficient mechanism, not a necessary characterization. Eckhardt explicitly leaves the broader CCAP problem open and asks whether the reduced free product of $(M_2 \otimes C[0, 1], \phi)$ with itself has the CCAP.

References

- [1] E. Ricard and Q. Xu, *Khintchine type inequalities for reduced free products and applications*, J. Reine Angew. Math. 599 (2006), 27–59; arXiv:math/0505302.
- [2] C. Eckhardt, *Free products and the lack of state-preserving approximations of nuclear C^* -algebras*, Proc. Amer. Math. Soc. 141 (2013), 2719–2727; arXiv:1011.2452; doi:10.1090/S0002-9939-2013-11702-8.